Chapter 3: Solving Linear Rational Expectation Models

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1 Overview

This chapter summarizes methods to solve linear stochastic difference equations in discrete time, using some of the concepts reviewed in the first chapter. Many macroeconomics models can be cast in this form via first-order approximation of the non-linear equations of the model, so these methods have broad applicability. The prominence of linear stochastic difference equations is largely explained by the fact that they allow us to use the powerful tools of linear algebra. This makes computation of the models and estimation relatively straightforward. But before describing the solution methods, it is worth asking: What do we mean by a solution?

1.1 What is a Solution to a Model?

Macro models, which are the objects of these lectures, are typically composed of two things: exogenous shocks, e.g. ϵ_t, and a set of endogenous variables, such as inflation, output, and interest rate. A full description of a model relates all the endogenous and exogenous variables together via a set of equations, and then describes the stochastic process that the exogenous shocks follow. A solution to the model, then, tells us how the endogenous variables can be expressed as a function *only* of the exogenous processes and, possibly, the set of initial conditions. The main complication is that the models often involve forward-looking expectation variables. Below, we outline how a solution is obtained. The main focus is on analytic solutions.

1.2 Two Motivating Examples

Consider the following two time series models:

$$y_t = \phi y_{t-1} + \epsilon_t \tag{1}$$

$$E_t y_{t+1} = \phi y_t + \epsilon_t \tag{2}$$

where ϕ is a coefficient and E_t is an expectation operator. The shock ϵ_t is a mean zero *iid* exogenous shock. The variable y_t is the endogenous variable. Equation (1) is a backward-looking equation while (2) is a forward-looking equation, i.e. it involves an expectation variable. When we refer to solving a model, we mean that we want to express the endogenous variable y_t as a function of the exogenous variable ϵ_t and, possibly, an initial condition. We will now solve each model.

Solving backwards

Since equation (1) holds at time t, it also holds at time $t-1$, i.e. $y_{t-1} = \phi y_{t-2} + \epsilon_{t-1}$. Substituting this into the original equation yields: $y_t = \phi^2 y_{t-2} + \phi\epsilon_{t-1} + \epsilon_t$. Now, substitute for $y_{t-2} = \phi y_{t-3} + \epsilon_{t-2}$ and so on to obtain:

$$y_t = \phi^t y_0 + \sum_{s=0}^{t-1} \phi^s \epsilon_{t-s}, \tag{3}$$

where y_0 is an *initial condition.* For this solution to be *bounded* we need $|\phi| < 1$, otherwise the first term $\phi^t y_0$ will grow progressively larger over time and the model will explodes. Hence for there to be a *bounded* solution then $|\phi| < 1$. The equation (3) the represents the *unique bounded solution* of model (1), i.e. we have expressed the endogenous variable as a function of only the *exogenous shock* and *an initial condition.*

Solving forwards

Write (2):

$$y_t = \frac{1}{\phi} E_t y_{t+1} - \frac{1}{\phi}\epsilon_t.$$

Since this holds at time t, it also holds at time $t+1$, yielding: $y_{t+1} = \frac{1}{\phi} E_{t+1} y_{t+2} - \frac{1}{\phi}\epsilon_{t+1}$. Now take expectation of both sides to yield: $E_t y_{t+1} = \frac{1}{\phi} E_t E_{t+1} y_{t+2} - \frac{1}{\phi} E_t \epsilon_{t+1}$. The key observation is that, by the *Law of Iterated Expectations*, we can write: $E_t E_{t+1} y_{t+2} = E_t y_{t+2}$, so that $E_t y_{t+1} = \frac{1}{\phi} E_t y_{t+2}$. Substitute this into the equation above to yield: $y_t = \frac{1}{\phi}\frac{1}{\phi} E_t y_{t+2} - \frac{1}{\phi}\epsilon_t$. We can repeat this for $E_t y_{t+2}$ and $E_t y_{t+3}$ and so on to obtain:

$$y_t = (\frac{1}{\phi})^T E_t y_{t+T} - \frac{1}{\phi}\epsilon_t.$$

The condition for y_t to have a *unique bounded* solution is that the first term, $(\frac{1}{\phi})^T E_t y_{t+T}$, disappears as T increases, i.e.

$$\lim_{T\to\infty} (\frac{1}{\phi})^T E_t y_{t+T} = 0,$$

which will be case if $|\phi| > 1$ and $\{\epsilon_t\}$ is bounded. When $|\phi| > 1$, we can thus write (2) as:

$$y_t = -\frac{1}{\phi}\epsilon_t \tag{4}$$

and $|\phi|$ >1 is thus critical for there to be *a unique bounded solution* for y_t. Equation (4) is the solution of model (2), i.e. we have expressed the *endogenous variable* as a function only of the *exogenous shock.*

Generalization: Preview

To summarize, for a unique bounded solution we need $|\phi| < 1$ for the backward-looking model, while we need $|\phi| > 1$ for the forward-looking model. The more general condition, which holds when y_t is a vector of variables and ϕ is a matrix, is that we need the eigenvalues of the matrix ϕ to be inside the unit circle for backward-looking models, while they need to be *outside of the unit circle* in case of forward-looking models (see discussion below).

Even more generally, when the model has both variables q_t, which enter as expectation variables i.e. $E_t q_{t+1}$, and variables k_t, which enter with a lag i.e. k_{t-1}, then we need as many eigenvalues of a properly defined matrix (see below) outside the unit circle as there are expectation variables, and as many eigenvalues inside the unit circle as there are variables that enter with a lag. In order to show this, which is known as the method of Blanchard and Kahn, we need to introduce matrix notation and some concepts from linear algebra that we briefly review below. It is important to recognize, however, that the logic of Blanchard and Kahn is *no more complicated* than the logic of the two examples above, as we will see now.

1.3 Review: Some Helpful Concepts from Linear Algebra

Here, I review some concepts from matrix algebra. It is very brief, and the idea is that you use this as keywords to explore in better depth yourself, if warranted.

Determinant of a matrix

The determinant of a matrix A, denoted $|\mathbf{A}|$, is a scalar that can be computed for any square matrix and encodes useful properties of it (for example,

we will use it to compute eigenvalues, see below).

Example:

The determinant of a 2×2 matrix is $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$.

Eigenvalues

Eigenvalues of a $n \times n$ matrix A are the roots $P(\lambda)$ that solve the determinant of $|A - \lambda I| = P(\lambda)$, where I is the *identity matrix*, e.g. for $2 \cdot 2$ it is $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$. It is useful to remember the following properties of the eigenvalues:

1. The sum of the eigenvalues equals the trace of A, i.e. the sum of the diagonal elements of A: $tr(A) = \sum_{i=1}^{n} a_{ii} = \sum_{i=1}^{n} \lambda_i$.
2. The product of the eigenvalues equals the determinant of A: $det(A) = \lambda_1 \cdot \lambda_2 \cdot ... \cdot \lambda_n$.
3. The eigenvalues of A^k, for k positive integer, are λ_1^k, λ_2^k, ..., λ_n^k.
4. Singular matrices have $\lambda = 0$.
5. If A is invertible (hence all the eigenvalues are nonzero), then the eigenvalues of A^{-1} are λ_1^{-1}, λ_2^{-1}, ..., λ_n^{-1}.
6. Diagonal matrices have the eigenvalues on their diagonal.

Example:

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

In order to find the eigenvalues, we first define:

$$\begin{aligned} P(\lambda) &= |A - \lambda I| \\ &= \left| \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right| \\ &= \begin{vmatrix} 2-\lambda & 1 \\ 1 & 2-\lambda \end{vmatrix} = (2-\lambda)^2 - 1 \\ &= \lambda^2 - 4\lambda + 3 \end{aligned}$$

Then, we set $P(\lambda) = \lambda^2 - 4\lambda + 3 = 0$ and we find the roots. The eigenvalues are: $\lambda_1 = 1$ and $\lambda_2 = 3$. Note that:

1. $tr(A) = 2 + 2 = 4 = 1 + 3 = \lambda_1 + \lambda_2$;

2. $det(A) = 4 - 1 = 3 = 1 \cdot 3 = \lambda_1 \cdot \lambda_2$.

Imaginary Numbers

There is no guarantee that the roots of $P(\lambda)$ are real. Imaginary numbers (named so in a derogatory way by Descartes, because he thought they were useless!) guarantee that we can always find the roots of polynomials. An imaginary number is defined by $i^2 = -1$ and a complex number is written as $a + bi$, where a and b are real numbers. We say that a complex root is *outside the unit circle* if it is greater than one in absolute value, defined as $|a + bi| = \sqrt{a^2 + b^2}$; *inside the unit circle* if it is less than one. Remark: When complex roots show up in models, the impulse responses typically exhibit waves. While complex roots do show up in models, the *solutions are of course always real.* Essentially, complex roots are a device that helps us perform the certain matrix manipulation in the general settings when the matrices do not have real eigenvalues.

Eigenvectors

Usually the term eigenvector refers to the right eigenvector defined by the equation:

$$Ax_R = \lambda x_R.$$

The Blanchard and Kahn method reviewed below, however, uses the somewhat less common left eigenvector, defined as:

$$x_L A = \lambda x_L.$$

Example:
Consider the eigenvector of the matrix A in last example. It is defined as:

$$\begin{bmatrix} e_1 & e_2 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} = \lambda \begin{bmatrix} e_1 \\ e_2 \end{bmatrix}$$

Note that this defines two equations in two unknowns (e_1 and e_2). Consider $\lambda = 1$. Then, the first equations yields $2e_1 + e_2 = e_1$ and the second equation yields $e_1 + 2e_2 = e_2$, i.e. both equations yield $e_1 = -e_2$. Hence, if we normalize $e_2 = 1$, then the left eigenvector is:

$$\begin{bmatrix} e_1 & e_2 \end{bmatrix}_{\lambda=1} = \begin{bmatrix} -1 & 1 \end{bmatrix}$$

A useful decomposition

Suppose a $n \times n$ matrix A has n distinct eigenvalues. Collect these into a matrix $\Lambda = \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & .. & 0 \\ 0 & 0 & \lambda_n \end{bmatrix}$. Collect the right eigenvectors into an $n \times n$ matrix $T = [\mathbf{x}_1, \mathbf{x}_2, ...\mathbf{x}_n]$. Then:

$$AT = [A\mathbf{x}_1, A\mathbf{x}_2, ...A\mathbf{x}_n]$$

and, since these are eigenvectors, then, by definition:

$$AT = [\lambda_1\mathbf{x}_1, \lambda_2\mathbf{x}_2, ...\lambda_n\mathbf{x}_n]$$

so that:

$$[\lambda_1\mathbf{x}_1, \lambda_2\mathbf{x}_2, ...\lambda_n\mathbf{x}_n] = [\mathbf{x}_1, \mathbf{x}_2, ...\mathbf{x}_n] \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & .. & 0 \\ 0 & 0 & \lambda_n \end{bmatrix} = T\Lambda.$$

Thus, we have that:

$$AT = T\Lambda.$$

Since the eigenvalues are distinct (by assumption, we consider generalization below), then the eigenvectors are known to be *linearly independent*, meaning that $|T| \neq 0$ and T^{-1} exists. It follows that we can post-multiply the formula above by T^{-1} so that we obtain:

$$A = T\Lambda T^{-1},$$

which is a useful decomposition of A as we now will see below. One useful observation, which we use in the next subsection, is that it gives useful way to understand the value of A in a higher power. In particular we can show that:

$$A^j = T\Lambda^j T^{-1}$$

To see this, write $A^2 = T\Lambda T^{-1} \times T\Lambda T^{-1} = T \times \Lambda \times (T^{-1}T) \times \Lambda \times T^{-1} = T \times \Lambda \times (I_n) \times \Lambda \times T^{-1} = T\Lambda^2 T^{-1}$, where I_n is the identity matrix. You can similarly do this for A^3 and so on for any j in A^j.

Jacobian decomposition

Often matrices will have repeated eigenvalues, but in the derivation above we assumed that they were all distinct, which made T invertible. If this is not the case, it means that the eigenvectors are not linearly independent and thus we cannot go through the steps above, which involved inverting the matrices. For this reason, we generally use instead a Jacobian decomposition which avoids this problem (see e.g. Hamilton), i.e. it guarantees that the matrices are invertible.

Schur decomposition

While the Jacobian decomposition has nice theoretical properties, it is found to sometimes lead to numerical errors when implemented numerically. The Schur decomposition and some variants work better and are typically found in computer codes, the details of which I will omit.

2 The Two Motivation Examples Generalized using Linear Algebra

Backward looking example

The backward looking example (1) may seem special. In fact, it applies to all backward looking systems that can be of arbitrary dimension. Consider the AR(p) process:

$$y_t = \phi_1 y_{t-1} + \phi_2 y_{t-2} + \phi_3 y_{t-3} + ... + \phi_p y_{t-p} + \epsilon_t.$$

This model can be written as:

$$\xi_t = F\xi_{t-1} + v_t,$$

where $\xi_t = \begin{bmatrix} y_t \\ y_{t-1} \\ ... \\ y_{t-p} \end{bmatrix}, v_t = \begin{bmatrix} \epsilon_t \\ 0 \\ ... \\ 0 \end{bmatrix}, F = \begin{bmatrix} \phi_1 & \phi_2 & ... & \phi_p & 0 \\ 1 & 0 & ... & 0 & 0 \\ .. & .. & ... & .. & .. \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}.$

This system can now be solved recursively exactly like before, i.e. $\xi_1 = F\xi_0 + v_1$ and $\xi_2 = F\xi_1 + v_2$ so that $\xi_2 = F^2\xi_0 + Fv_1 + v_2$. This can be repeated until data t to yield:

$$\xi_t = F^t\xi_0 + F^{t-1}v_1 + F^{t-2}v_2 + ... + Fv_{t-1} + v_t \tag{5}$$

To simplify, let us assume that F has distinct eigenvalues and stack the right eigenvectors of each eigenvalue into the matrix T. Then we can write:

$$F = T\Lambda T^{-1},$$

where $\Lambda = \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & ... & 0 \\ 0 & 0 & \lambda_p \end{bmatrix}$, with λ_i being the eigenvalues, and $T = \begin{bmatrix} e_1^1 & .. & e_p^1 \\ .. & ... & .. \\ e_1^p & .. & e_p^p \end{bmatrix}$, with the first row demoting the eigenvector associated with the first eigenvalue and so on. This implies that:

$$F^t = T\Lambda^t T^{-1}.$$

Thus, while before we needed the parameter $|\phi| < 1$ so that $\lim_{t\to\infty} \phi^t = 0$, we now need that the eigenvalues of F are inside the unit circle so that:

$$\lim_{t\to\infty} \Lambda^t = \begin{bmatrix} \lim_{T\to\infty}(\lambda_1)^t & 0 & 0 \\ 0 & ... & 0 \\ 0 & 0 & \lim_{t\to\infty}(\lambda_p)^t \end{bmatrix} = \mathbf{0}.$$

Note that for the more general case that accounts for the possibility of repeated eigenvalues we need to use Jordan decomposition (the Schur decomposition tends to be better, however, numerically).

Forward looking example

Similar intuition can be developed with a forward looking system, with the eigenvalues now needing to be outside the unit circle. I leave this as an exercise (when you see this in any text, it typically means that the instructor started doing it but found too tedious to complete before class).

3 Follow-up remark: What is a solution?

I already stressed that we consider a solution to a model an expression that relates the endogenous variables to the "primitive" shocks of the model and initial conditions. Many mathematical programs and papers report a "solution," however, by relating the endogenous variables to lagged variables of themselves and the shocks. The lagged variables are called "state variables" of the model, they are known at time t and the resulting expression often has useful interpretation. It should be well understood, however, that it is a straightforward step to express a solution of this form in terms of the primitive shocks using (5). The Slutsky equation in Chapter 1, for example, is already of this form, as is any ARMA(p,q) model for that matter. When deriving the solution using the method of Blanchard and Khan, we will leave it of this form and thus you may often see in this class a system referred to as a "solution" even if strictly speaking we have only left it as describing how the endogenous variables are related to the shocks and possibly endogenous state variables.

4 Solving General Linear Rational Expectation Models using method of Blanchard and Kahn

We now state a general system that involves both lagged and expectation variables and show how it can be solved using the method of Blanchard and Kahn (1980) [1].

General system

$$AE_tY_{t+1} = BY_t + C\epsilon_t$$

where A, B, C are matrices. Y_t and ϵ_t are $NY \times 1$ and $NX \times 1$ vectors, while A and B are $NY \times NY$ matrices and C is $NY \times NX$.

Y_t can be partitioned into $Y_t = \begin{bmatrix} q_t \\ k_{t-1} \end{bmatrix}$, with k_{t-1} being a $NK \times 1$ (where $0 \leq NK \leq NY$). The vector k_{t-1} contains all elements known at time $t-1$ and earlier, hence we have $E_tY_{t+1} = \begin{bmatrix} Eq_{t+1} \\ k_t \end{bmatrix}$, i.e the variables contained in q_t are variables that enter with expectations. We call the variables in k_{t-1} predetermined at time t.

Simplified form

Assuming A is invertible (not needed in general but simplifies the exposition), we can pre-multiply the left and right-hand sides by the inverse of A and rewrite the system as:

$$E_tY_{t+1} = \begin{bmatrix} E_tq_{t+1} \\ k_t \end{bmatrix} = \begin{bmatrix} m_{11} & m_{12} \\ m_{21} & m_{21} \end{bmatrix} \begin{bmatrix} q_t \\ k_{t-1} \end{bmatrix} + v_t, \tag{6}$$

where we call the $NY \times NY$ matrix containing the $m's$ M, which is equal to $A^{-1}B$, while v_t is equal to $A^{-1}C\epsilon_t$ ($NY \times 1$ vector). Let us call the eigenvalues that are outside of the unit circle "unstable". Denote by NUE the number of eigenvalues of M that are unstable. Then, Blanchard and Kahn prove the following:

Proposition 1. *1. If $NUE = NY - NK$, i.e. there are as many unstable eigenvalues as non-predetermined variables, then for each k_0 and bounded $\{\epsilon_t\}$ there exists a unique and bounded solution $\{Y_t\}$ that satisfies the system for $t > 0$.*

2. if $NUE < NY - NK$, i.e. there are less unstable eigenvalues than non-predetermined variables, then for each k_0 and bounded $\{\epsilon_t\}$ there exists infinitely many bounded $\{Y_t\}$ satisfying the system for $t > 0$.

3. If $NUE > NY - NK$, i.e. there are more unstable eigenvalues than non-predetermined variables, then there will be generically no bounded $\{Y_t\}$ satisfying the system, given k_0 and bounded $\{\epsilon_t\}$ for $t > 0$.

Example:

Consider our example

$$E_t y_{t+1} = \phi y_t + \epsilon_t \tag{7}$$

Here, the matrix M=ϕ has the eigenvalue $\lambda = \phi$. NY=1 and NK=0. Hence, for unique bounded solution NUE=1 and Blanchard and Kahn's theorem says that the system has a unique bounded solution if $\phi > 1$, as we have already confirmed.

Example
Consider the standard New Keynesian model (that we will study in detail later):

$$\begin{aligned}
\pi_t &= \kappa Y_t + \beta E_t \pi_{t+1} \\
Y_t &= E_t Y_{t+t} - \sigma(i_t - E_t \pi_{t+1} - r_t^n) \\
i_t &= \rho i_{t-1} + \phi_\pi \pi_t + \epsilon_t
\end{aligned}$$

You can put it in the form:

$$A E_t Y_{t+1} = B Y_t + C \epsilon_t$$

as follows:

$$\begin{bmatrix} \beta & 0 & 0 \\ \sigma & 1 & -\sigma \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} E_t \pi_{t+1} \\ E_t Y_{t+1} \\ i_t \end{bmatrix} = \begin{bmatrix} 1 & -\kappa & 0 \\ 0 & 1 & 0 \\ \phi_\pi & 0 & \rho \end{bmatrix} \begin{bmatrix} \pi_t \\ Y_t \\ i_{t-1} \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ -\sigma & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} r_t^n \\ \epsilon_t \end{bmatrix}$$

Here $NY = 3$, $NK = 1$, $NX = 2$. You can feed this system (i.e. the matrices A, B, and C) directly into some matlab program and get a solution of the form that expresses Y_t, π_t, and i_t as a function of the shocks and state variable i_{t-1}.

Observe that, in practice, most programs express a "solution" simply as an equation relating the variables in question to the shocks and lagged variables of the endogenous variables (with the understanding that you can then easily express the variables as a function of the shocks only by repeated substitution).

4.1 Explicit solution that shows Blanchard and Khan's theorem

Solving the system explicitly is useful for you to learn how to derive closed form solutions. It's also helpful as an informal proof for the Blanchard and Kahn theorem. Let me note that this may be difficult to follow, the best way to proceed might be to read this together with examples provided below, which makes the method more concrete. Consider the following system, using the notation from before:

$$\begin{bmatrix} E_t q_{t+1} \\ k_t \end{bmatrix} = \begin{bmatrix} m_{11} & m_{12} \\ m_{21} & m_{21} \end{bmatrix} \begin{bmatrix} q_t \\ k_{t-1} \end{bmatrix} + v_t \tag{8}$$

The definition of eigenvalue λ_i and its associated eigenvector e_i implies that:

$$e_i M = \lambda_i e_i, \tag{9}$$

where e_i is a $1 \times NY$ vector. Using this definition, we can rewrite the system by first pre-multiplying by e_i and then using (9) to obtain:

$$e_i \begin{bmatrix} E_t q_{t+1} \\ k_t \end{bmatrix} = e_i M \begin{bmatrix} q_t \\ k_{t-1} \end{bmatrix} + e_i v_t = \lambda_i e_i \begin{bmatrix} q_t \\ k_{t-1} \end{bmatrix} + e_i v_t$$

where the crucial step is that we use (9) to replace $e_i M = \lambda_i e_i$. Let us now define a variable that is a linear combination of the endogenous variables as follows: $\xi_t = e_i \begin{bmatrix} q_t \\ k_{t-1} \end{bmatrix}$. Using this definition, we can rewrite the system as:

$$E_t \xi_{t+1} = \lambda_i \xi_t + e_i v_t$$

but note that this is a simple univariate as (2)! It can thus be written as:

$$\xi_t = \frac{1}{\lambda_i} E_t \xi_{t+1} - \frac{1}{\lambda_i} e_i v_t = (\frac{1}{\lambda_i})[\frac{1}{\lambda_i} E_t \xi_{t+2} - \frac{1}{\lambda_i} e_i E_t v_{t+1}] - \frac{1}{\lambda_i} e_i v_t =$$

$$= ... = (\frac{1}{\lambda_i})^j E_t \xi_{t+j} + w_t^i$$

where $w_t^i \equiv -\frac{1}{\lambda_i} e_i v_t$, which has a unique bounded solution since, if $|\lambda_i| > 1$, then $lim_{j \to \infty} (\frac{1}{\lambda_i})^j E_t \xi_{t+j} = 0$.

Hence, for every unstable λ_i, we have the linear relationship:

$$\xi_t = \begin{bmatrix} e_1^i & e_2^i \end{bmatrix} \begin{bmatrix} q_t \\ k_{t-1} \end{bmatrix} = w_t^i$$

where the $1 \times NY$ eigenvector e_i is partitioned into two components of dimension $1 \times (NY - NK)$ and $1 \times NK$ respectively. Note that we can rewrite the expression as:

$$e_1^i q_t = -e_2^i k_{t-1} + w_t^i$$

We need as many unstable eigenvalues as the number of variables in q_t. Thus, for every unstable eigenvalues i=1,2,3,..., we have a relationship as above. Suppose that there are as many unstable eigenvalues as the dimension of q_t (there are $NY - NK$ eigenvalues outside the unit circle). Then, we can construct two matrices, Q_1 and Q_2 composed as follows: Q_1 is a matrix $(NY - NK) \times$

$(NY-NK)$, whose rows are the first partition of the corresponding $(NY-NK)$ eigenvectors, e_1^i, for $i = 1, 2, ..., NY-NK$, each with dimension $1 \times (NY-NK)$; Q_2 is a matrix of dimension $(NY - NK) \times NK$, whose rows are the second partition of the corresponding $(NY-NK)$ eigenvectors, e_2^i, for $i = 1, 2, ..., NY-NK$, each with dimension $1 \times NK$. Formally, we can write:

$$\begin{bmatrix} e_1^1 \\ ... \\ e_1^{NY-NK} \end{bmatrix} q_t = Q_1 q_t = \begin{bmatrix} e_2^1 \\ ... \\ e_2^{NY-NK} \end{bmatrix} k_{t-1} + w_t = Q_2 k_{t-1} + w_t,$$

where w_t is the vector containing all the w_t^i, hence of dimension $(NY-NK) \times 1$. Finally, we can pre-multiply both sides by the inverse of Q_1 and solve for q_t:

$$q_t = Q_1^{-1} Q_2 k_{t-1} + Q_1^{-1} w_t.$$

On the other hand, the second equation of (8) gives:

$$k_t = m_{21} q_t + m_{22} k_{t-1} + a_2 \epsilon_t,$$

where we can substitute for the solution of q_t by the former equation. These two equations describe the solution, where we have expressed the endogenous variables as a function of the shocks and their own lagged values. A solution in terms of the fundamental shocks is now straightforward to obtain via recursive substitution.

4.2 Using method of undermined coefficients to obtain closed form solutions

In practice, to obtain analytic solution, it is sometimes useful to:

1. Check that the Blanchard and Kahn conditions are satisfied.

2. If they are, then you now there is a unique bounded solution of the form:

$$q_t = a k_{t-1} + b v_t$$

$$k_t = c k_{t-1} + d v_t$$

where a,b,c,d are unknown coefficients. You can substitute this conjectured solution into the original system and "match coefficients" in order to get a solution. This is best explained via examples, see below. Uhlig (1998) has a nice toolkit that shows Blanchard and Kahn generally using the method of undetermined coefficients.

4.3 Example 1

Consider the following equation (which will show up later in the course), where β, κ, λ and x^* are coefficients:

$$\beta E_t \phi_{t+1} - (1 + \beta + \kappa^2/\lambda)\phi_t + \phi_{t-1} = \kappa x^*$$

How is this solved? We can use the method of Blanchard and Kahn. To simplify, let us write this in terms of deviation from steady state, i.e. $\bar{\phi} = -x^*\lambda/\kappa$:

$$\beta E_t(\phi_{t+1} - \bar{\phi}) - (1 + \beta + \kappa^2/\lambda)(\phi_t - \bar{\phi}) + (\phi_{t-1} - \bar{\phi}) = 0$$

or:

$$E_t\hat{\phi}_{t+1} - (1 + \beta + \kappa^2/\lambda)\beta^{-1}\hat{\phi}_t + \beta^{-1}\hat{\phi}_{t-1} = 0,$$

using the notation that hat denotes deviation from steady state.

Let's write this in the form of Blanchard and Kahn's proposition:

$$\begin{bmatrix} E_t\hat{\phi}_{t+1} \\ \hat{\phi}_t \end{bmatrix} = \begin{bmatrix} (1 + \beta + \kappa^2/\lambda)\beta^{-1} & -\beta^{-1} \\ 1 & 0 \end{bmatrix} \begin{bmatrix} \hat{\phi}_t \\ \hat{\phi}_{t-1} \end{bmatrix} = M \begin{bmatrix} \hat{\phi}_t \\ \hat{\phi}_{t-1} \end{bmatrix}$$

Now, we need one eigenvalue of M to be greater than one and one to be less than one in absolute value. The characteristic equation is:

$$((1 + \beta + \kappa^2/\lambda) - \mu)(-\mu) + \beta^{-1} = 0$$

or

$$\beta\mu^2 - (1 + \beta + \kappa^2/\lambda)\mu + 1 = 0$$

This equation has one root greater than one and one less than one, hence this linear equation has a unique bounded solution.

Solving using left eigenvectors

Recall that the left eigenvector associated with the unstable eigenvalue is $eM = \mu_2 e$, where $\mu_2 > 1$. We use this to rewrite the system as:

$$e \begin{bmatrix} E_t\hat{\phi}_{t+1} \\ \hat{\phi}_t \end{bmatrix} = e \begin{bmatrix} (1 + \beta + \kappa^2/\lambda)\beta^{-1} & -\beta^{-1} \\ 1 & 0 \end{bmatrix} \begin{bmatrix} \hat{\phi}_t \\ \hat{\phi}_{t-1} \end{bmatrix} = \mu_2 e \begin{bmatrix} \hat{\phi}_t \\ \hat{\phi}_{t-1} \end{bmatrix}$$

Define $\Gamma_t \equiv e \begin{bmatrix} \hat{\phi}_t \\ \hat{\phi}_{t-1} \end{bmatrix}$, so that this implies that:

$$\Gamma_t = \frac{1}{\mu_2} E_t\Gamma_{t+1} = \frac{1}{\mu_2}\frac{1}{\mu_2} E_t\Gamma_{t+2} = ... = (\frac{1}{\mu_2})^T E_t\Gamma_{t+T} = \lim_{T\to\infty} (\frac{1}{\mu_2})^T E_t\Gamma_{t+T} = 0$$

for a bounded Γ_T, since $\mu_2 > 1$. Let's find e:

$$\begin{bmatrix} e_1 & 1 \end{bmatrix} \begin{bmatrix} (1+\beta+\kappa^2/\lambda)\beta^{-1} & -\beta^{-1} \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} \mu_2 e_1 & \mu_2 \end{bmatrix}$$

$$\begin{bmatrix} e_1(1+\beta+\kappa^2/\lambda)\beta^{-1}+1 & -e_1\beta^{-1} \end{bmatrix} = \begin{bmatrix} \mu_2 e_1 & \mu_2 \end{bmatrix}$$

i.e. $e_1 = -\beta\mu_2$

Our solution is $\Gamma_t = 0$ and hence:

$$\begin{bmatrix} e_1 & 1 \end{bmatrix} \begin{bmatrix} \hat{\phi}_t \\ \hat{\phi}_{t-1} \end{bmatrix} = -\beta\mu_2\hat{\phi}_t + \hat{\phi}_{t-1} = 0$$

or:

$$\hat{\phi}_t = \beta^{-1}\mu_2^{-1}\hat{\phi}_{t-1}$$

Using methods of undermined coefficients

In practice, when I show that the number of eigenvalues is right to ensure a unique bounded solution, I typically use the method of undermined coefficients to find a closed form solution. My experience is that this is a bit quicker. In this example, given that there is one unstable eigenvalue and so we have a unique bounded solution, we know that the solution is of this form: $\hat{\phi}_t = \phi^r\hat{\phi}_{t-1}$, but we don't know ϕ^r. Substitute this into:

$$E_t\hat{\phi}_{t+1} - (1+\beta+\kappa^2/\lambda)\beta^{-1}\hat{\phi}_t + \beta^{-1}\hat{\phi}_{t-1} = 0$$

to obtain:

$$\phi^r\phi^r\phi_{t-1} - (1+\beta+\kappa^2/\lambda)\beta^{-1}\phi^r\phi_{t-1} + \beta^{-1}\hat{\phi}_{t-1} = 0$$

$$[\phi^r\phi^r - (1+\beta+\kappa^2/\lambda)\beta^{-1}\phi^r + \beta^{-1}]\hat{\phi}_{t-1} = 0$$

Hence, ϕ^r solves the quadratic equation: $(\phi^r)^2 - (1+\beta+\kappa^2/\lambda)\beta^{-1})\phi^r + \beta^{-1} = 0$.

You will notice that this is of course simply the "characteristic" equation. Hence, here we have: $\phi^r = \mu_1$. Thus, the solution is:

$$\hat{\phi}_t = \mu_1\hat{\phi}_{t-1}$$

By the way, notice that $\mu_1 = \beta^{-1}\mu_2^{-1}$, by our earlier solution. Indeed, the determinant of the matrix M is equal to β^{-1} and, by the properties of the eigenvalues, we know that: $det(M) = \mu_1 \cdot \mu_2$.

4.4 Example 2

Consider the following system of equations:

$$\pi_t = \kappa Y_t + \pi_{t-1} + u_t$$
$$Y_t = E_t Y_{t+1} + \gamma \pi_{t-1} + \sigma r_t^e$$

How do you set this up in the form that we have studied so that Blanchard and Kahn can be applied? Usually you just stick the equations into a computer and hit enter. Notice that the "predetermined" variables are all those that are already determined at time t.

$$\begin{bmatrix} E_t Y_{t+1} \\ \pi_t \end{bmatrix} = \begin{bmatrix} 1 & -\gamma \\ \kappa & 1 \end{bmatrix} \begin{bmatrix} Y_t \\ \pi_{t-1} \end{bmatrix} + \begin{bmatrix} -\sigma & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} r_t^e \\ u_t \end{bmatrix}$$
$$\begin{bmatrix} E_t Y_{t+1} \\ \pi_t \end{bmatrix} = M \begin{bmatrix} Y_t \\ \pi_{t-1} \end{bmatrix} + \begin{bmatrix} -\sigma & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} r_t^e \\ u_t \end{bmatrix}$$

So, in order to obtain a unique bounded solution you need the matrix M to have one unstable eigenvalue, i.e. one eigenvalue that is greater than one in absolute value and one that is less than one. The eigenvalues of M solve:

$$(1-\mu)^2 - \gamma\kappa = \mu^2 - 2\mu + 1 - \gamma\kappa = 0$$

Are there values of γ and κ that yield a unique bounded solution? If you have a larger system, you just expand the vectors of endogenous variables.

4.5 Example 3

More to be added

5 References

[1] Olivier Jean Blanchard and Charles M. Kahn. "The solution of linear difference models under rational expectations". *Econometrica: Journal of the Econometric Society* (1980), pp. 1305–1311.

[2] Harald Uhlig. "A toolkit for analyzing nonlinear dynamic stochastic models easily". In: *Computational Methods for the Study of Dynamic Economies*. Ed. by Ramon Marimon and Andrew Scott. Oxford University Press, 1998. Chap. 3.

6 Problems

Problem 1
Consider this model:

$$x_t = \rho E_t x_{t+1} + y_{t-1} + \epsilon_t$$
$$y_t = \gamma x_t + u_t$$

where u_t and ϵ_t are iid with zero mean and bounded support, E_t is an expectation operator and the parameters ρ, $\gamma > 0$.

i. Using method of Blanchard and Kahn, under what conditions about the parameters ρ and γ is there a unique bounded rational expectation equilibrium in this model? *(**Hint:** You should be able to show that if a particular inequality is satisfied, then there is a unique bounded solution).*

ii. Show an impulse response function for x_t and y_t to an innovation in ϵ_t and u_t.